



Part 2:

Transient Analysis

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Elements: c and l

Energy-storing elements

Symbulator can simulate two types of energy-storing elements: capacitors and inductors. Below is what you need to know on how to describe them.

Describing a capacitor

In Symbulator, an ideal **capacitor** is described as follows: *first* the name of the capacitor, which must start with the letter **c**, coma; *second* the name of the first node of the capacitor, another coma; *third* the name of the second node of the capacitor, another coma; and *fourth* the value of the capacitor (in F). For transient analysis, a fifth term is expected: the initial condition of the capacitor, given in volts, as voltage in 1st node minus voltage in 2nd node.

Notice that, in the calculator, variables c1 through c99 are reserved. So you cannot use them as names or as values for a capacitor. Instead, what I recommend is that you use letters (e.g. **ca**, **cb**) or double c (e.g. **cc1**, **cc2**).

For example, a capacitor called **ca**, connected between nodes **1** and **2** with a value of **1F**, and an initial condition of **0V**, would be described thus: **ca,1,2,1,0** Since the value of capacitors is often given in milli or micro farads, Symbulator will interpret any **m** or **μ** in the value of a capacitor as an indication that the values are in mF or μF respectively. Other permissible SI units are **k** for kilo, **n** for nano and **p** for pico.

Simulation answers

After a TR simulation, Symbulator stores a series of answers for the capacitor, labeled with easy to remember names for your convenience.

- The *voltage of each of its nodes* with reference to the ground is stored in a variable called **v** and the name of the node.
- The *voltage drop in the capacitor*, e.g. the voltage in the 1st node minus the voltage in the 2nd node, is stored in a variable called **v** and the name of the capacitor. For example, for a capacitor called **ca**, the voltage drop is in **vca**.
- The *current through the capacitor*, flowing from the first node towards the second, is stored in a variable called **i** and the name of the capacitor. For example, the current flowing through capacitor **ca**, from the first node towards the second, is stored in **ica**.

Note on capacitors in DC

In a DC simulation, the current through a capacitor is defined as 0.

This is how capacitors are described in Symbulator.

*Variables **c1** through **c99** are reserved in the calculator.*

*Symbulator interprets any **m** or **μ** in the value of the capacitor to mean milli or micro, respectively.*

Describing an inductor

This is how inductors are described in Symbulator.

*Symbulator interprets any **m** or **μ** in the value of the inductor to mean milli or micro, respectively.*

In Symbulator, an ideal **inductor** is described as follows: *first* the name of the inductor, which must start with the letter **I**, coma; *second* the name of the first node of the inductor, another coma; *third* the name of the second node of the inductor, another coma; and *fourth* the value of the inductor (in H). For transient analysis, a fifth term is expected: the initial condition of the inductor, given in amperes, as current from 1st node towards 2nd node.

For example, an inductor called **I1**, connected between nodes **1** and **2** with a value of **1H**, and an initial condition of **0A**, would be described thus: **I1,1,2,1,0**. Since the value of inductors is often given in mili or micro henrys, Symbulator will interpret any **m** or **μ** in the value of an inductor as an indication that the values are in mH or μH respectively. Other permissible SI units are **k** for kilo, **n** for nano and **p** for pico.

Simulation answers

After a TR simulation, Symbulator stores a series of answers for the inductor, labeled with easy to remember names for your convenience.

- The *voltage of each of its nodes* with reference to the ground is stored in a variable called **v** and the name of the node.
- The *voltage drop in the inductor*, e.g. the voltage in the 1st node minus the voltage in the 2nd node, is stored in a variable called **v** and the name of the inductor. For example, for an inductor called **I1**, the voltage drop is in **vl1**.
- The *current through the inductor*, flowing from the first node towards the second, is stored in a variable called **i** and the name of the inductor. For example, the current flowing through inductor **I1**, from the first node towards the second, is stored in **il1**.

Note on inductors in DC

In a DC simulation, the voltage drop in an inductor is defined as 0.

Transient simulation

***s\tr** is the gate to run a transient simulation*

You will learn the subtleties of running transient simulations in Symbulator through a series of examples presented below. For now, suffice it to say that to run a transient simulation in Symbulator, we use an access program, or "gate", called **s\tr**, which takes one argument: the circuit description in string form. Below you will find plenty of examples of TR simulations, showing you how to describe sources that are step, impulse and function of time, as well as sources that are constant, both numeric and symbolic.

For transient simulations, Symbulator uses a software for Laplace transforms called DiffEq, by Lars Frederiksen. Make sure you have it properly installed!

Ok, let's get right into it!

Solved Problems

The following examples are taken from *Elementary Linear Circuit Analysis* (2ed) by Leonard S. Bobrow. Moving forward, I refer to this book as **Bo2**.

Bo2's Example 4.15

EXAMPLE 4.15

For the circuit shown in Fig. 4.37(a), let us determine v given that i is described by Fig. 4.37(b). In this case, let us consider three intervals of time.

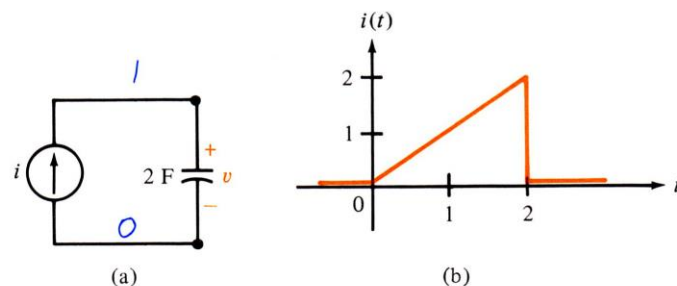


Fig. 4.37 (a) Circuit with a capacitor. (b) Applied current.

Solution. For $t \leq 0\text{s}$, since the current source is inactive, there is nothing going on. Without doing a simulation, we know the voltage v is **0** volts.

For $0\text{s} < t \leq 2\text{s}$, the source is active. By inspection of the graph (b), we learn that for this interval the value of the source is t . That is to say, the current in amperes is the same as the time in seconds. We describe the circuit and run the transient analysis simulation using the command shown below:

```
s\tr("j,0,1,t;c,1,0,2,0",0)
```

Here, `s\tr` is the transient analysis gate and the circuit to be simulated is provided between the quotes. The description of the source has four terms: name (starting with `j`), first node, second node, and value. In this case the value is t . The description of the capacitor has five terms: name (starting with `c`), first node, second node, value in F, and the capacitor's initial condition in volts. In this case, the initial condition of the capacitor is **0** volts.

Once the simulation is complete, ask for the value of the voltage: `vc`. We get the value shown below, which is correct.

$$\frac{t^2}{4}$$

For $2\text{s} < t$, the voltage is equal to whatever value it had at time 2s :

$$vc|t=2$$

We get a value of **1** volt, which is correct.



A word on source values

For the purpose of a transient analysis, Symbolator accepts many types of source values. The only thing you must do is describe these values properly. Below are examples on how to describe four different types of source values.

Step values: symbolic

A source with an unknown step value starting in $t=0$ is described using a variable and the step function $u(t)$. For example, a voltage source $e1$, connected between nodes 1 and 0, with a value of V volts starting in time = 0 seconds, would be described as follows:

$e1,1,0,V*u(t)$

Step values: numerical

A step source with a known numerical value can also be described as above:

$e1,1,0,12*u(t)$

However, to save some typing, when your value is numerical, you can skip the $u(t)$. Symbolator will assume it is a numerical step value you are passing.

$e1,1,0,12$

The results you get are the same. So, if you know the numerical value, you can save yourself some typing by skipping the $u(t)$ in the value description.

Impulse values

A source with an impulse value in time $t=0$ is described using its value, be it symbolic or numerical, and the step function $\delta(t)$. For example, a current source $j1$, connected between nodes 0 and 1, with an impulse value of i amperes in time = 0 seconds, would be described as follows:

$j1,0,1,i*\delta(t)$

Values as functions of time

A source with a value that is a function of time, such as a ramp value, or a sinusoidal or exponential value, can be described by writing the value as an expression in terms of variable t . An example is found in the problem above.

$j,0,1,t$

Values as function of t will activate the Impala mode, in order to save time.

Dependent values

Sources with dependent values are described as we have seen before, e.g.

$j,0,1,3*vr1$

Bo2's Example 5.1

EXAMPLE 5.1

Let us begin this example by considering the circuit in Fig. 5.4, in which there is a switch that is closed for time $t < 0$, is opened at time $t = 0$, and stays open for all time $t > 0$. For this circuit, let us determine $v_C(t)$ for all values of t .

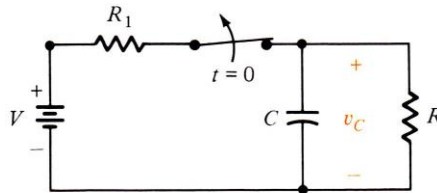


Fig. 5.4 RC circuit with a switch.

Solution. Here we have two intervals. For the first interval, when $t < 0$ s, we assume that things have been steady for a long time now and any transient effect has long passed. So for this first interval of $t < 0$ s, we do a DC simulation to find the initial conditions for the second interval.

```
s\dc("e,1,0,V:r1,1,2,r1:c,2,0,c:r,2,0,r")
```

When the simulation is *Done*, we ask for v_C , the voltage in the capacitor **c**. We get the expression shown below, which is correct.

$$\frac{r v}{r + r1}$$

For the second interval, when $0s \leq t$, we do a transient simulation. We give the capacitor as initial condition its voltage from the previous interval.

```
s\tr("c,2,0,c,r*v/(r+r1):r,2,0,r",0)
```

When the simulation is *Done*, we ask for v_C again. We get the expression below, which is correct.

$$e^{-t/cr} \frac{r v}{r + r1}$$

A word on intervals

For transient analyses in Symbolator, every time a switch opens or closes marks the end of a time interval and the beginning of another time interval. A simulation can only be done for one time interval. When we run a TR simulation and get answers in terms of t , this variable t must be understood as the time elapsed since the start of that interval in particular. This is a distinction without a difference as long as the interval we simulate with TR starts at $t=0$, which is most often the case. However, sometimes problems have switch changes in times other than $t=0$. A first example is given in page 212, where a switch closes at $t=1$ second. In these cases, it is important to remember that Symbolator's answers for a TR simulation are given in terms of t , where t means the time elapsed since the beginning of that interval.

Bo2's Drill Exercise 5.1

DRILL EXERCISE 5.1

For the circuit shown in Fig. DE5.1, find a single expression for all t for (a) $v_C(t)$ and (b) $i_C(t)$.

Answer: (a) $6 - 6(1 - e^{-4t})u(t)$ V; (b) $-2e^{-4t}$ A

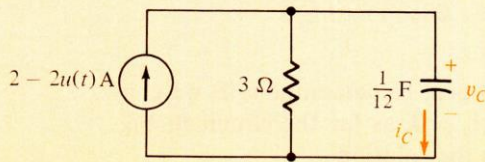


Fig. DE5.1

Solution. Again, we have two intervals: the first for DC, the second for TR.

For the first interval, when $t < 0$ s, the current source is 2A. Simulate in DC.

```
s\dc("j,0,1,2:r,1,0,3:c,1,0,1/12"):{vc,ic}
{ 6 , 0 }
```

So the voltage in the capacitor is 6V. We store this value in a variable called $\mathbf{vc_0}$, to use it as initial condition of the capacitor for the second interval.

```
vc>vc0
```

For the second interval, when $0 \leq t$, we do a transient simulation. The current source is 0A, and the capacitor has an initial condition: voltage $\mathbf{vc_0}$.

```
s\tr("j,0,1,0:r,1,0,3:c,1,0,1/12,vc0",0):{vc,ic}
```

When the simulation is *Done*, we get the expressions below. They are right.

```
{ 6e^{-4t} , -2e^{-4t} }
```

Bo2's p224 5.2

EXAMPLE 5.2

For the circuit shown in Fig. 5.7, let us determine $v_C(t)$, $i_C(t)$, and $v(t)$ for all time.

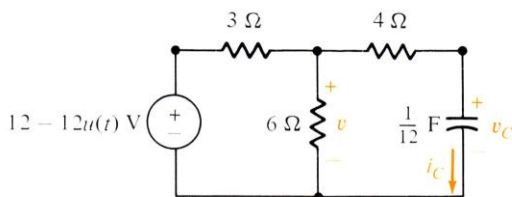


Fig. 5.7 Circuit for Example 5.2.

Solution. Again, we have two intervals: the first for DC, the second for TR.

For the interval when $t < 0$ s, the voltage source is 12A. Simulate in DC.

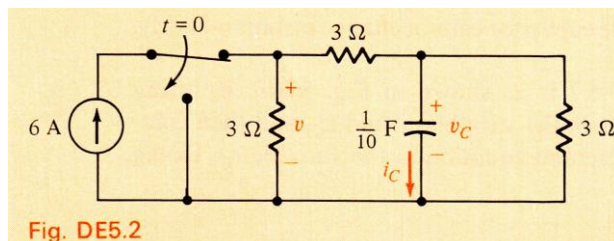
```
s\dc("e,1,0,12:r3,1,2,3:r6,2,0,6:r4,2,3,4:c,3,0,1/12"):{vc,ic,v2}
{ 8 , 0 , 8 }
```

For the interval when $0s \leq t$, the voltage source is 0V. Simulate in TR, giving the capacitor as initial condition its voltage from the previous interval.

```
s\tr("e,1,0,0:r3,1,2,3:r6,2,0,6:r4,2,3,4:c,3,0,1/12,8",0):{vc,ic,v2}
```

$$\left\{ 8e^{-2t}, \frac{4}{3}e^{-2t}, \frac{8}{3}e^{-2t} \right\}$$

Bo2's Drill Exercise 5.2



DRILL EXERCISE 5.2

For the circuit shown in Fig. DE5.2, find an expression for all t for (a) $v_C(t)$, (b) $i_C(t)$, and (c) $v(t)$.

Answer: (a) $6 - 6(1 - e^{-5t})u(t)$ V; (b) $-3e^{-5t}u(t)$ A; (c) $12 - 12u(t) + 3e^{-5t}u(t)$ V

Solution. For $t < 0s$, the 6A source is part of the circuit. Simulate in DC.

```
s\dc("j,0,1,6:r10,1,0,3:r12,1,2,3:r20,2,0,3:c,2,0,1/10"): {vc,ic,v1}
```

$$\{ 6, 0, 12 \}$$

For $0s \leq t$, the source is no longer relevant and can be excluded from the circuit. Simulate in TR, giving the capacitor its proper initial condition.

```
s\tr("r10,1,0,3:r12,1,2,3:r20,2,0,3:c,2,0,1/10,6",0):{vc,ic,v1}
```

$$\{ 6e^{-5t}, -3e^{-5t}, 3e^{-5t} \}$$

Bo2's Figure 5.10a

Look at the simple resistor-inductor (RL) circuit shown below, where at time $t=0$ the inductor current is $i_L(0)$. Determine $v_L(t)$, $i_L(t)$ and $v_R(t)$ for $t \geq 0$.



Since we are already given the initial condition, we only run the transient simulation for $t \geq 0$. In our circuit description below, the initial condition is **il0**.

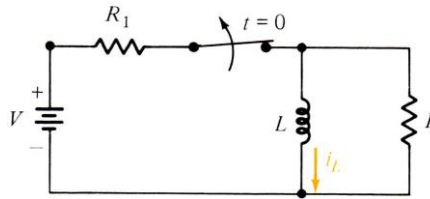
```
s\tr("l,1,0,l,il0:r,1,0,r",0):{vl,il,vr}
```

After 22 seconds, we get the right answers:

$$\left\{ -il0 r e^{\frac{-rt}{l}}, il0 e^{\frac{-rt}{l}}, -il0 r e^{\frac{-rt}{l}} \right\}$$

Bo2's Example 5.3

Determine $i_L(t)$ for all t .



Solution. For $t < 0s$, simulate in DC.

```
s\dc("e,1,0,v:r1,1,2,r1:l,2,0,l:r2,2,0,r"):il
```

$$\frac{v}{r1}$$

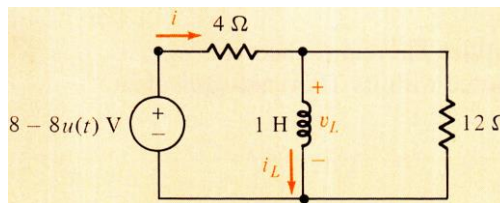
For $t \geq 0s$, simulate in TR, giving the inductor its initial condition.

```
s\tr("l,2,0,l,v/r1:r2,2,0,r",0):il
```

$$e^{-\frac{r}{L}t} \frac{v}{r1}$$

Bo2's Drill Exercise 5.3

Find $i_L(t)$, $v_L(t)$ and $i(t)$ for all t .



Solution. For $t < 0s$, simulate in DC.

```
s\dc("e,1,0,8:r4,1,2,4:l,2,0,1:r12,2,0,12"): {il,vl,ir1}
```

$$\{2, 0, 2\}$$

For $t \geq 0s$, simulate in TR, giving the inductor its initial condition of 2A.

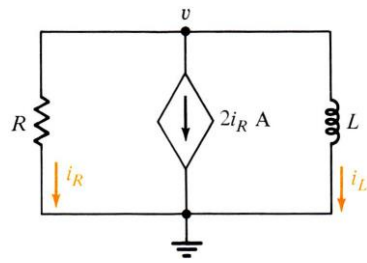
```
s\tr("e,1,0,0:r4,1,2,4:l,2,0,1,2:r12,2,0,12",0): {il,vl,ir4}
```

$$\{2e^{-3t}, -6e^{-3t}, \frac{3}{2}e^{-3t}\}$$

In my machine the simulation took 30 seconds, 12 seconds of these (40%) were used in finding the inverse Laplace of the answers. Later we will learn a trick to save time by specifying to Symbolator which answers are required.

Bo2's p230 (Dependent source)

Find $i_L(t)$ for $t \geq 0$, given that $i_L(0) = 5A$.



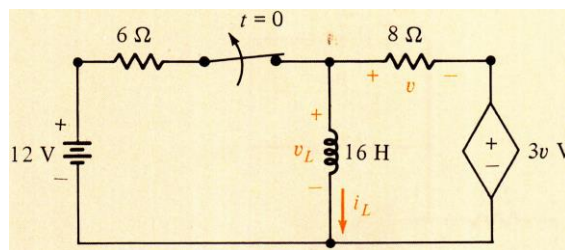
We only run the transient simulation for $t \geq 0$, with initial condition 5A.

```
s\tr("r,v,0,r:j,v,0,2ir:l,v,0,l,5",0):il
```

$$5e^{\left(\frac{-R}{3L}t\right)}$$

Bo2's Drill Exercise 5.4 (Dependent source)

Find $i_L(t)$ and $v_L(t)$ for all t .



Solution. For $t < 0$ s, simulate in DC. This gives us the initial condition.

```
s\dc("ei,1,0,12:r6,1,2,6:l,2,0,16:r8,2,3,8:ed,3,0,3vr8"): {il,vl}
```

$$\{2, 0\}$$

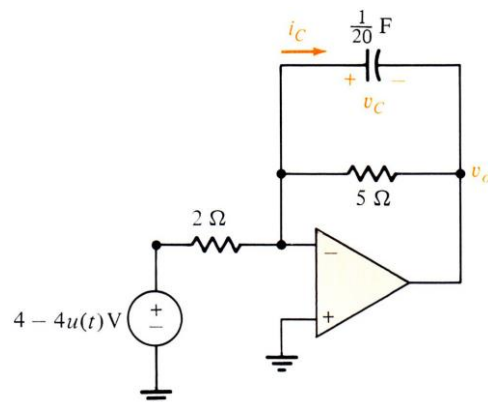
For $t \geq 0$ s, simulate in TR, giving the inductor its initial condition.

```
s\tr("l,2,0,16,2:r8,2,3,8:ed,3,0,3vr8",0): {il,vl}
```

$$\{2e^{-2t}, -64e^{-2t}\}$$

Bo2's Example 5.5 (Op Amp)

Determine $v_C(t)$, $i_C(t)$ and $v_o(t)$ for all t .



Solution. For $t < 0s$, simulate in DC. This gives us the initial condition.

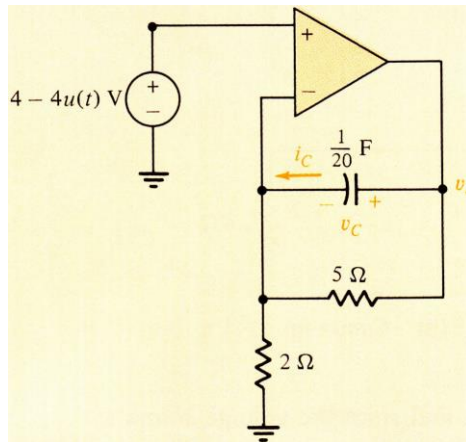
```
s\dc("e,3,0,4:r2,3,1,2:r5,1,o,5:c,1,o,1/20:o,0,1,o"):{vc,ic,vo}
{ 10 , 0 , -10 }
```

For $t \geq 0s$, simulate in TR, giving the capacitor its initial condition.

```
s\tr("r2,0,1,2:r5,1,o,5:c,1,o,1/20,10:o,0,1,o",0):{vc,ic,vo}
{ 10 e^{-4t} , -2 e^{-4t} , -10 e^{-4t} }
```

Bo2's Drill Exercise 5.5 (Op Amp)

Determine $v_c(t)$, $i_c(t)$ and $v_o(t)$ for all t .



Solution. For $t < 0s$, simulate in DC. This gives us the initial condition.

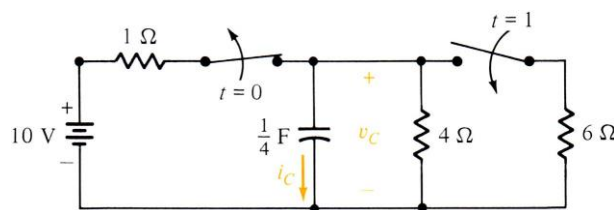
```
s\dc("e,2,0,4:o,2,1,o:c,o,1,1/20:r5,o,1,5:r2,1,0,2"):{vc,ic,vo}
{ 10 , 0 , 14 }
```

For $t \geq 0s$, simulate in TR, giving the capacitor its initial condition.

```
s\tr("o,0,1,o:c,o,1,1/20,10:r5,o,1,5:r2,1,0,2",0):{vc,ic,vo}
{ 10 e^{-4t} , -2 e^{-4t} , 10 e^{-4t} }
```

Bo2's Example 5.6

In the circuit shown below, there are two switches: one that opens at time $t=0$ and one that closes at time $t=1$ second. Determine $v_c(t)$ and $i_c(t)$ for all t .



As we mentioned before (in page 207), every time a switch moves marks the end of one interval and the beginning of another interval for Symbolator.

For the first interval, which corresponds to $t < 0$, we simulate in DC.

```
s\dc("e,2,0,10:r1,2,1,1:c,1,0,1/4:r4,1,0,4"):{vc,ic}
{8,0}
```

The voltage in the capacitor at the end of this interval will serve as the initial condition of the capacitor for the next interval. The second interval runs between 0 and 1 second, e.g. $0 < t \leq 1$ second. We simulate in TR.

```
s\tr("c,1,0,1/4,8:r4,1,0,4",0):{vc,ic}
{8e^{-t}, -2e^{-t}}
```

The voltage in the capacitor at the end of this second interval will serve as initial condition for the third interval. We can use its exact value, e.g. $8 \cdot e^{-1}$, but the textbook prefers using its approximate value, e.g. 2.943.

The third interval corresponds to $t > 1$ second. We simulate in TR. To make things easier for Symulator, we replace the resistors by their equivalent.

```
s\tr("c,1,0,1/4,2.943:re,1,0,p(4,6)",1)
```

Ask for $\{vc, ic\}$. The expressions we get are equivalent to:

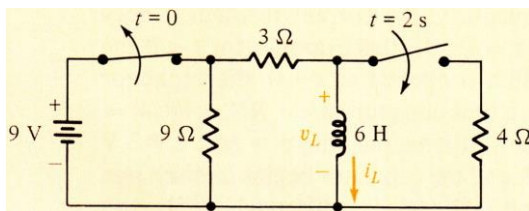
$$2.943 e^{-\frac{5(t-1)}{3}} \quad -1.226 e^{-\frac{5(t-1)}{3}}$$

These are the right answers, as can be seen by checking the book's answers.

$$v_C(t) = \begin{cases} 8 \text{ V} & \text{for } t < 0 \\ 8e^{-t} \text{ V} & \text{for } 0 \leq t < 1 \text{ s} \\ 2.943e^{-5(t-1)/3} \text{ V} & \text{for } t \geq 1 \text{ s} \end{cases} \quad i_C(t) = \begin{cases} 0 \text{ A} & \text{for } t < 0 \\ -2e^{-t} \text{ A} & \text{for } 0 \leq t < 1 \text{ s} \\ -1.226e^{-5(t-1)/3} \text{ A} & \text{for } t \geq 1 \text{ s} \end{cases}$$

Bo2's Drill Exercise 5.6

Let's see another example where a TR simulation is done for an interval that starts at a time other than $t=0$. Determine $i_L(t)$ and $v_L(t)$ for all t .



For the first interval, which corresponds to $t < 0$, we simulate in DC.

```
s\dc("e,1,0,9:r9,1,0,9:r3,1,2,3:l,2,0,6"):{il,vl}
{3,0}
```

The 3A current in the inductor is its initial condition for the second interval, which goes from 0 to 2 second, e.g. $0 < t \leq 2$ seconds. We simulate in TR.

```
s\tr("r,1,0,9+3:l,1,0,6,3",0):{il,vl}
```

$$\{ 3 e^{-2t}, -36 e^{-2t} \}$$

The current in the capacitor at the end of this second interval will serve as initial condition for the third interval. Find its approximate value thus:

$$i_L|_{t=2}.$$

$$.055$$

The third interval corresponds to $t > 2$ second. We simulate in TR. To make things easier for Symbolator, we replace the resistors by their equivalent.

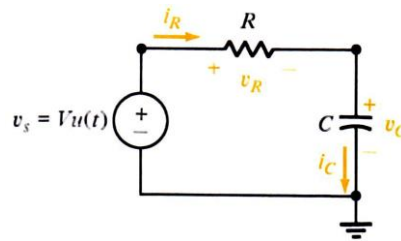
$$s\text{tr}("r,1,0,p(9+3,4):l,1,0,6,.055",2):\{i_l,v_l\}$$

$$.055 e^{\left(\frac{-(t-2)}{2}\right)} \quad -.165 e^{\left(\frac{-(t-2)}{2}\right)}$$

These are the kind of expressions your book or professor are looking for.

Bo2's Figure 5.19

In the following circuit, assume that all the initial conditions are zero. The source is a step function $u(t)$ with value V volts. Find i_R , v_R , i_C and v_C .



As discussed in page 206, a step source with symbolic value requires that we use the $u(t)$ nomenclature to describe it. Since the value is V volts, we describe the source as $V*u(t)$.

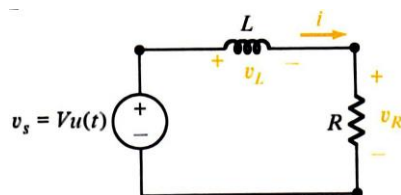
$$s\text{tr}("e,1,0,V*u(t):r,1,2,r:c,2,0,c,0",0):\{v_c,v_r,i_c\}$$

There is no need to ask for i_R , since $i_R = i_C$. We get the following expressions:

$$\{ v - v e^{\left(\frac{-t}{cr}\right)}, v e^{\left(\frac{-t}{cr}\right)}, \frac{v}{r} e^{\left(\frac{-t}{cr}\right)} \}$$

Bo2's Figure 5.24

In the following circuit, assume that all the initial conditions are zero. The source is a step function $u(t)$ with value V volts. Find i_R , v_R , i_L and v_L .



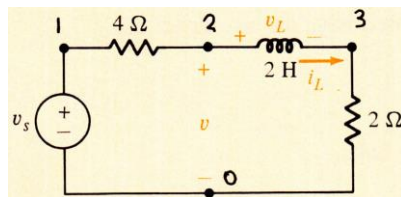
$$s\text{tr}("e,1,0,V*u(t):l,1,2,l,0:r,2,0,r",0):\{v_l,v_r,i_l\}$$

There is no need to ask for i_R , since in a series circuit it will be identical to i_L .

$$\{v e^{\frac{-t}{\tau}}, v(1 - e^{\frac{-t}{\tau}}), \frac{v}{r}(1 - e^{\frac{-t}{\tau}})\}$$

Bo2's Drill Exercise 5.7

For the circuit below, with a step source of 12 volts, find i_L , v_L , i_L and v .



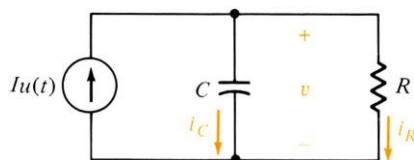
As discussed in page 206, a step source with numerical value can be described without the $u(t)$ nomenclature. Thus, we describe the source as 12.

$$\text{s\textbackslash tr('e,1,0,12:r4,1,2,4:l,2,3,2,0:r2,3,0,2',0):\{i_l,v_l,v_2\}}$$

$$\{2 - 2e^{-3t}, 12e^{-3t}, 8e^{-3t} + 4\}$$

Bo2's Figure 5.26

In the following circuit, assume that all the initial conditions are zero. The source is a step function $u(t)$ with value I amperes. Find i_C , i_R , and v .



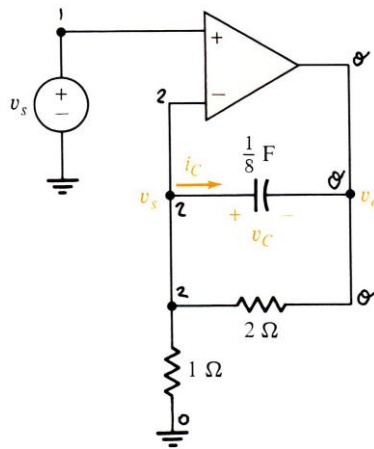
As discussed in page 206, a step source with symbolic value requires that we use the $u(t)$ nomenclature to describe it. Since the value is I amperes, we describe the source as $I*u(t)$. Thus:

$$\text{s\textbackslash tr('j,0,1,i*u(t):c,1,0,c,0:r,1,0,r',0):\{v_1,i_r,i_c\}}$$

$$\{(i - i e^{\frac{t}{cr}})*r, i - i e^{\frac{t}{cr}}, i e^{\frac{t}{cr}}\}$$

Bo2's Example 5.7 (Op Amp)

Suppose that $v_s(t)=u(t)$. Find v_C , i_C , and v_o .

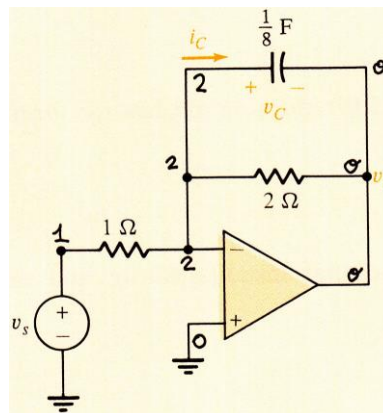


This source is a step source with a value of 1 volt. Since its value is numerical, the $u(t)$ nomenclature is not needed. We describe the source plainly as 1.

```
s\tr("e,1,0,1:o,1,2,o:c,2,o,1/8,0:r2,2,o,2:r1,2,0,1",0):{vc,ic,vo}
{ 2 e^{-4t} - 2, -e^{-4t}, 3 - 2e^{-4t} }
```

Bo2's Drill Exercise 5.8 (Op Amp)

Suppose that $v_s(t)=u(t)$. Find v_C , i_C , and v_o .

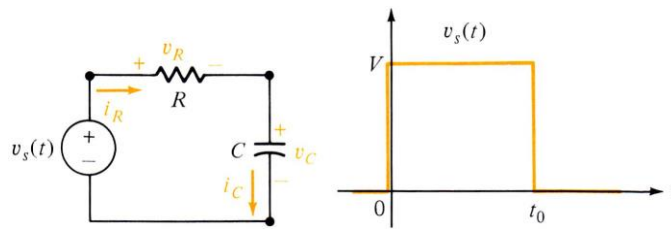


Again, following the same logic as above, we describe the source plainly as 1.

```
s\tr("e,1,0,1:r1,1,2,1:r2,2,o,2:c,2,o,1/8,0:o,0,2,o",0):{vc,ic,vo}
{ 2 - 2 e^{-4t}, e^{-4t}, 2 e^{-4t} - 2 }
```

Bo2's p249 F5.28

For the circuit left, the source is $v_s(t) = V u(t) - V u(t-t_0)$. This is shown in the plot to the right. Given this, find the voltage drop in the capacitor for all t .



Because the source has two steps, this problem has to be analyzed in two intervals. For the first interval, the initial conditions are zero and the source value is $V \cdot u(t)$. Our circuit description for the first interval is given below.

$$\text{s\texttt{tr}("e,1,0,V*u(t):r,1,2,r:c,2,0,c,0",0):vc}$$

The expression for the capacitor's voltage drop in the first interval is:

$$v - v e^{\left(\frac{-t}{c \cdot r}\right)}$$

The second interval starts at t_0 . In it, the initial condition of the capacitor is given by the value of the expression above when $t=t_0$. Here is how we find it:

$$vc|_{t=t_0}$$

$$v - v e^{\left(\frac{-t_0}{c \cdot r}\right)}$$

The circuit description for the second interval uses the expression above as the initial condition of the capacitor. In this second interval, the source has a value of 0 volts, which is another way to say that it becomes a short, so there is no need to include it in the circuit description.

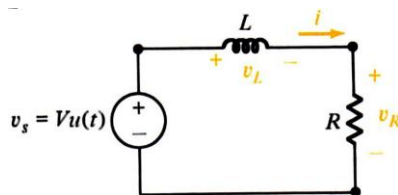
$$\text{s\texttt{tr}("r,0,2,r:c,2,0,c,v-e^{\left(\frac{-t_0}{c \cdot r}\right)}*v",t_0):vc}$$

$$\left(v - v e^{\left(\frac{-t_0}{c \cdot r}\right)}\right) e^{\left(\frac{-t+t_0}{c \cdot r}\right)}$$

This is the expression for the capacitor's voltage drop in the second interval.

Bo2's Drill Exercise 5.9

Find the current through the inductor if the source is $v_s(t) = V u(t) - V u(t-t_0)$.



Because the source has two steps, this problem has to be analyzed in two intervals. For the first interval, the initial conditions are zero and the source value is $V \cdot u(t)$. Our circuit description for the first interval is given below.

$$\text{s\texttt{tr}("e,1,0,V*u(t):l,1,2,l,0:r,2,0,r",0):il}$$

The expression for the inductor's current in the first interval is:

$$\frac{v}{r} \left(1 - e^{\left(\frac{-r}{l} t \right)} \right)$$

The second interval starts at t_0 . In it, the initial condition of the inductor is given by the value of the expression above when $t=t_0$. Here is how we find it:

$$i|_{t=t_0}$$

$$\frac{v}{r} \left(1 - e^{\left(\frac{-t_0 r}{l} \right)} \right)$$

The circuit description for the second interval uses the expression above as the initial condition of the inductor. In this second interval, the source becomes a short, so there is no need to include it in the circuit description.

$$s\backslash tr("l,0,2,l,v/r-e^{(-t_0*r/l)}*v/r:r,2,0,r",0):il$$

$$\frac{v}{r} \left(1 - e^{\left(\frac{-t_0 r}{l} \right)} \right) e^{\left(\frac{-r}{l} t \right)}$$

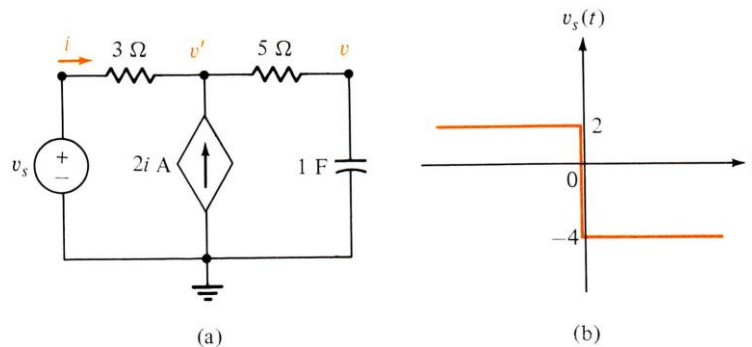
To put this in terms of the same t as the first interval, we replace t with $t-t_0$:

$$\frac{v}{r} \left(1 - e^{\left(\frac{-t_0 r}{l} \right)} \right) e^{\left(\frac{-r}{l} (t-t_0) \right)}$$

This is the expression for the inductor's current in the second interval.

Bo2's Figure 5.36

For the circuit shown in (a), find the voltage drop in the capacitor, given that the voltage source is as shown in (b).



This problem requires two intervals. In the first interval, a DC analysis of the circuit with a source of 2 V gives us the voltage in the capacitor.

$$s\backslash dc("e,1,0,2:r3,1,2,3:r5,2,3,5:c,3,0,1:j,0,2,2ir3"):vc$$

2

This is the initial condition for the capacitor in the second interval, which we analyze using TR and a source value of -4 V.

$$s\backslash tr("e,1,0,-4:r3,1,2,3:r5,2,3,5:c,3,0,1,2:j,0,2,2ir3",0):vc$$

$$6e^{\left(\frac{-t}{6} \right)} - 4$$

This is the expression for the voltage drop in the capacitor after $t=0$.

The only tool

Unless you indicate otherwise, a Symbolator simulation will give you a whole array of answers: voltages in all nodes, and voltage drops, currents and power consumed in all elements, etc. When the circuit you are solving has complicated symbolic expressions, storing all these answers in the memory requires time. In case you don't want all the answers, but only a chosen few, you *may* save some time by using the `s\only` tool to tell Symbolator – before the simulation – which answers you want to save. Find it in Symbolator's custom menu. The argument of `s\only` is either an empty variable or a string with a variable or a list of variables separated by commas, as shown below.

`s\only(ir1)`

`s\only("ir1")`

`s\only("ir1,va,vr2")`

The `s\only` tool is valuable in TR analysis, because you can save time by not having Symbolator find the inverse Laplace of answers that are not needed. Let's see how to use the `s\only` tool using a variation of a familiar example.

*If you are only
interested in a few
answers, using the
tool `s\only` can
save you some time*

Bo2's Drill Exercise 5.11

For the circuit of Bo2's Example 5.7, change the value of the capacitor to $\frac{1}{4}$ F. Find $v_c(t)$ for the case that the source is $v_s(t)=1$ V for $t<0$ and 3V for $t\geq 0$.

We let Symbolator know that we are only interested in one variable: v_c .

`s\only("vc")`

Then we run the simulation for the first interval, just as we did before, but with the new capacitor and source values.

`s\dc("e,1,0,1:o,1,2,o:c,2,o,1/4:r2,2,o,2:r1,2,0,1"):vc`

-2

This is the initial condition for the next interval. Before we run the simulation for the second interval, we let Symbolator know, again, that we are only interested in one variable: v_c .

`s\only("vc")`

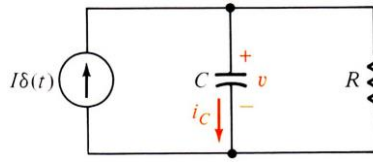
Then we run the simulation for the second interval, just as we did before, but with the new capacitor and source values.

`s\tr("e,1,0,3:o,1,2,o:c,2,o,1/4,-2:r2,2,o,2:r1,2,0,1",0):vc`

$4e^{(-2t)} - 6$

Bo2's Figure 5.38 (Impulse)

Find the voltage drop and current in the capacitor, given a source of $I \delta(t)$ A, where $\delta(t)$ is the impulse function and I is a constant.



As we mentioned back in page 206, impulse sources must be described using the $\delta(t)$ nomenclature. To save some typing time, you find it in the menu.

```
s\tr("j,0,1,i*\delta(t):c,1,0,c,0:r,1,0,r",0)
```

Once the simulation completes, we ask for the variables of interest:

vc

$$\left(\frac{i}{c}\right) e^{\left(\frac{-t}{cr}\right)}$$

ic

$$\left(\frac{-i}{cr}\right) e^{\left(\frac{-t}{cr}\right)}$$

Bo2's Drill Exercise 5.13 (Impulse)

For the circuit of Bo2's Example 5.7, find $v_c(t)$, $i_c(t)$ and $v_o(t)$ due to an impulse voltage source of $v_s(t) = \delta(t)$ (that is to say, a 1V impulse in $t=0$.)

Nothing new here. Again we use the $\delta(t)$ nomenclature for the source.

```
s\tr("e,1,0,\delta(t):o,1,2,o:c,2,o,1/8,0:r2,2,o,2:r1,2,0,1",0):{vc,ic,vo}
      { -8 e^{-4t}, 4 e^{-4t}, 8 e^{-4t} }
```

Bo2's Figure 5.39 (Ramp)

Find $i_L(t)$ and $v_L(t)$ for a circuit that has in parallel a resistor R , an inductor L and a current source with a ramp value $I r(t)$. Initial conditions are zero.

A ramp value $I r(t)$ means a value of $I t$. We describe this source simply as $i*t$.

```
s\tr("j,0,1,i*t:r,1,0,r:l,1,0,l,0",0):{il,vl}
      { \frac{il}{r} \left( e^{\left(\frac{-rt}{l}\right)} - 1 \right) + it, il \left( 1 - e^{\left(\frac{-rt}{l}\right)} \right) }
```

Bo2's Drill Exercise 5.14 (Ramp)

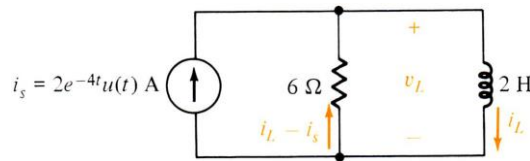
For the circuit of Bo2's Example 5.7, find $v_c(t)$ and $i_c(t)$ due to a ramp input voltage of $v_s(t) = r(t)$.

Nothing new here. The command below should be clear to you by now.

```
s\tr("e,1,0,t:o,1,2,o:c,2,o,1/8,0:r2,2,o,2:r1,2,0,1",0):{vc,ic}
      { \frac{1}{2} (1 - e^{-4t}) - 2t, \frac{1}{4} (e^{-4t} - 1) }
```

Bo2's Example 5.12 (Exponential)

Assume initial conditions zero. Find $i_L(t)$ and $v_L(t)$, given that $i_s(t) = 2 e^{-4t}$ A.



Describe the source as given to you by the problem, using an exponential.

$$\text{s\tr("j,0,1,2e^{(-4t):r,0,1,6:l,1,0,2,0",0):\{il,vl\}} \\ \{ 6 e^{-3t} - 6 e^{-4t}, 48 e^{-4t} - 36 e^{-3t} \}$$

Bo2's p265 (Exponential)

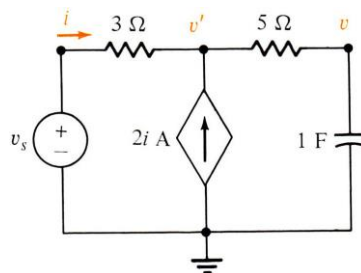
Repeat Bo2's Example 5.12 if the current source is now $i_s(t) = 2 e^{-3t}$ A.

All we do is change one digit in the circuit description: replace 4 with 3.

$$\text{s\tr("j,0,1,2e^{(-3t):r,0,1,6:l,1,0,2,0",0):\{il,vl\}} \\ \{ 6 t e^{-3t}, (12 - 36 t) e^{-3t} \}$$

Bo2's Example 5.13 (Exponential)

Find the voltage drop in the capacitor and the current in the 3Ω resistor, if the source value is $v_s(t) = 18e^{-t/2}$.



$$\text{s\tr("e,1,0,18e^{(-t/2):r3,1,2,3:r5,2,3,5:c,3,0,1,0:j,0,2,2ir3",0):\{vc,ir3\}} \\ \{ 9 e^{\left(\frac{-t}{6}\right)} - 9 e^{\left(\frac{-t}{2}\right)}, \frac{3}{2} e^{\left(\frac{-t}{2}\right)} - \frac{1}{2} e^{\left(\frac{-t}{6}\right)} \}$$

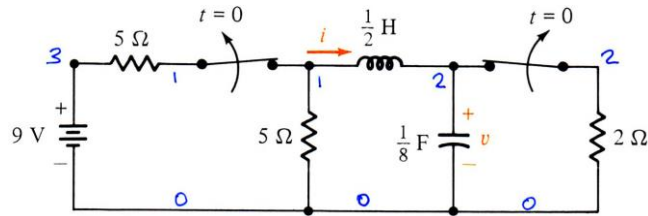
Bo2's Drill Exercise 5.16 (Exponential)

For the circuit of Bo2's Example 5.7, find $v_C(t)$ and $i_C(t)$ for the case when the source's value is $v_s(t) = 2 e^{-4t}$. As before, assume initial conditions are zero.

$$\text{s\tr("e,1,0,2e^{(-4t):o,1,2,o:c,2,o,1/8,0:r2,2,o,2:r1,2,0,1",0):\{vc,ic\}} \\ \{ -16 t e^{-4t}, (8 t - 2) e^{-4t} \}$$

Bo2's Example 6.1

Find the current in the inductor $i(t)$ and the voltage drop in the capacitor $v(t)$.



Since we have switches changing in time $t=0$, this is a two interval problem. The first interval is for $t<0$. We can analyze the first interval using DC.

```
s\dc("e,3,0,9:r2,3,1,5:r1,1,0,5:l,1,2,1/2:c,2,0,1/8:r3,2,0,2"): {vc,il}
{2,1}
```

These are the initial conditions for the second interval. The second is for $t\geq 0$. We can analyze the second interval using TR. We only care about **vc** and **il**.

```
s\only("vc,il"):s\tr("r1,1,0,5:l,1,2,1/2,1:c,2,0,1/8,2",0): {vc,il}
{ 4 e^{-2t} - 2 e^{-8t}, 2 e^{-8t} - e^{-2t} }
```

Notice that the use of the only tool saves approximately 10 seconds here.

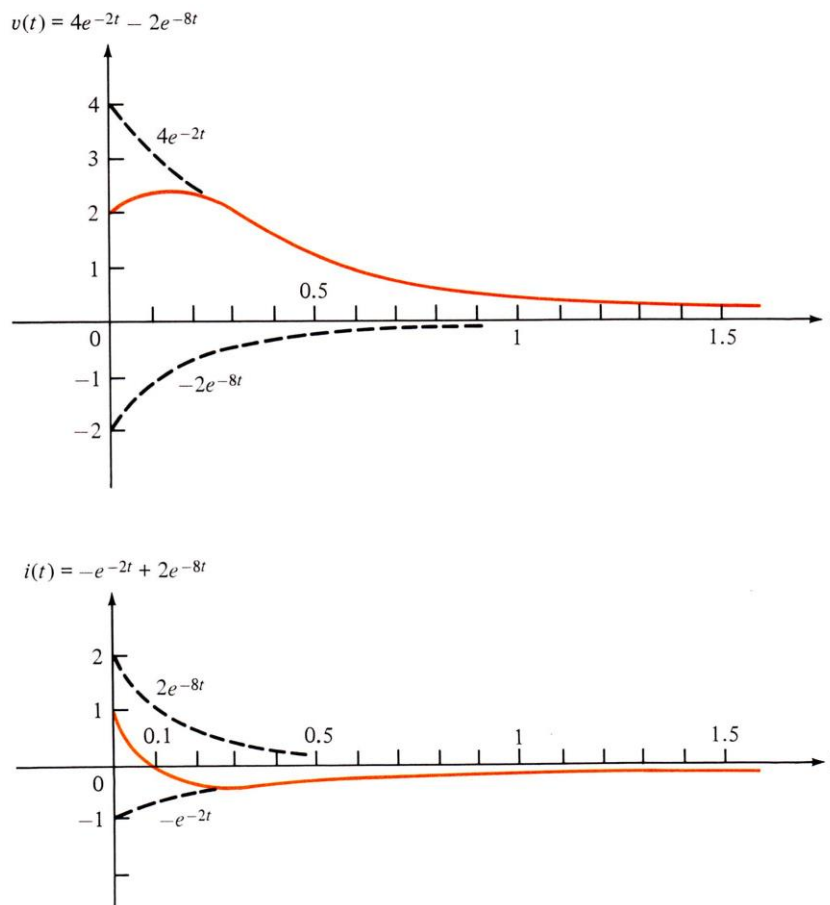
The plot tool

*Use the **s\plot** tool to plot functions of time*

Very often we are asked to plot functions of time. The calculator has extensive plotting capabilities, but just to save you some time, Symbolator includes a handy **plot** tool that allows you to plot functions of time. Imagine, for example, that in Bo2's Example 6.1 we are asked to plot $i(t)$ and $v(t)$ for times between 0 s and 1.5 s. The first step is to run the **plot** tool:

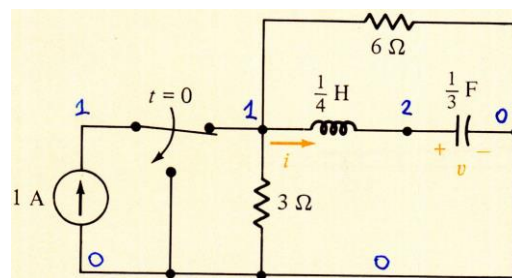
```
s\plot()
```

A window opens, asking you to enter three things: first, a function of time; second, a minimal time; and third, a maximal time. Let's plot $v_c(t)$ first. Unless you have deleted the value of the variable **vc**, you should have the answer to the simulation above stored in it. So enter **vc** as the function, **0** as the minimal time and **1.5** as the maximal time. Press Enter, and wait a little. The graph of the voltage drop in the capacitor between time 0 and 1.5 seconds should appear in the screen. Compare it to the graph given by the textbook, shown below. Repeat this procedure giving **il** as the function. Compare the resulting graph to that given by the textbook, shown below.



Bo2's Drill Exercise 6.1

For the circuit shown below, find $v(t)$ and $i(t)$ for $t \geq 0$.



Nothing new here. This is a two interval problem, as others seen before. The first interval, for $t < 0$, is analyzed in DC to find the initial conditions.

$$\text{s}\backslash\text{dc}(\text{"j,0,1,1:r3,1,0,3:r6,1,0,6:l,1,2,1/4:c,2,0,1/3"}):\{\text{vc,il}\}$$

$$\{2,0\}$$

These conditions are used in the TR analysis of the second interval, for $t \geq 0$.

$$\text{s}\backslash\text{only}(\text{"vc,il"}):\text{s}\backslash\text{tr}(\text{"r3,1,0,3:r6,1,0,6:l,1,2,1/4,0:c,2,0,1/3,2",0})$$

$$\text{vc}$$

$$3e^{-2t} - e^{-6t}$$

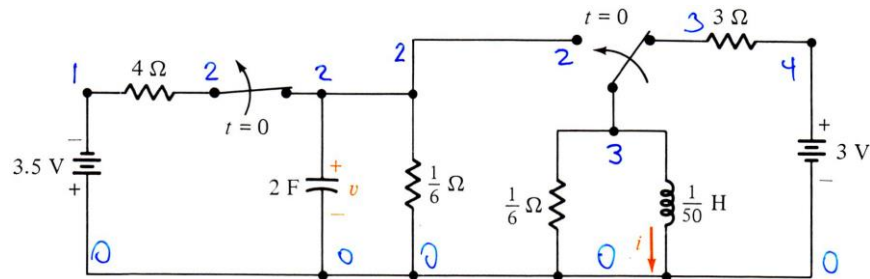
il

$$2e^{-6t} - 2e^{-2t}$$

Using **only** saves about 13 seconds, or about a third of the calculation time, because only about 8 seconds are used finding the inverse Laplace of the selected answers, instead of 21 seconds finding it for all the answers.

Bo2's Example 6.2

Find $i(t)$ and $v(t)$ for the circuit shown below.



I decided to simulate this using only fractions, not decimals. So I convert the 3.5V value to its exact fractional equivalent, thus:

exact(3.5)

$$7/2$$

Because of the switches, this is a two interval problem. The first interval is to be analyzed using DC. Notice that for $t < 0$ the circuit is such that we basically have two separate circuits. We simulate each one of them separately:

s\dc("e1,0,1,7/2:r1,1,2,4:c,2,0,2:r2,2,0,1/6"):vc

$$-7/50$$

s\dc("r3,3,0,1/6:l,3,0,1/50:r4,3,4,3:e2,4,0,3"):il

$$1$$

The second interval, for $t \geq 0$, is analyzed using TR. I could have described the circuit again from scratch. But out of laziness I preferred to copy/paste the descriptions from the DC simulations. To avoid renaming the nodes, I simulated the right switch as a short circuit between nodes 2 and 3.

s\only("vc,il"):s\tr("c,1,0,2,-7/50:r,1,0,p(1/6,1/6):l,1,0,1/50,1",0)

vc

$$\frac{-7}{50}e^{-3t} \cos(4t) - \frac{1}{50}e^{-3t} \sin(4t)$$

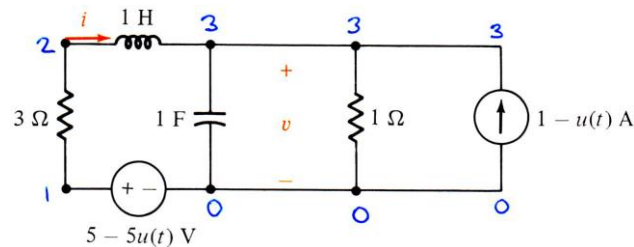
il

$$e^{-3t} \cos(4t) - e^{-3t} \sin(4t)$$

The TR simulation took 30 s in my calculator. These are the right answers.

Bo2's Example 6.3

Find $i(t)$ and $v(t)$ for the circuit shown below.



Since the sources change value, this is a two interval problem. For $t < 0$, in DC:

$$\text{s\texttt{dc}("e,1,0,5:r3,1,2,3:l,2,3,1:c,3,0,1:r1,3,0,1:j,0,3,1"):\{vc,il\}} \\ \{2,1\}$$

Using these as initial conditions, now we use TR for $t \geq 0$:

$$\text{s\texttt{only}("vc,il"):\texttt{s\texttt{tr}("r3,0,2,3:l,2,3,1,1:c,3,0,1,2:r1,3,0,1",0):\{vc,il\}} \\ \{(3t + 2)e^{-2t}, (1 - 3t)e^{-2t}\}$$

Is the textbook wrong?

Anybody that has even written a textbook, or read one for that matter, can attest to the fact that errors in the problems or the answers come with the territory. (As a matter of fact, I am sure this book I am writing right now has many errors.) We are all only human. Errors in an engineering textbook, however, can be *very* frustrating to an engineering student. When one is learning from a textbook, and the textbook has errors in the problems or answers, confusion and hair-pulling will likely ensue. If you think you have found an error in your circuit analysis textbook, Symbolator can help you bring some sanity to the situation to confirm whether your guess is right.

Below is an example of using Symbolator to confirm an error in a textbook.

Bo2's Drill Exercise 6.4

DRILL EXERCISE 6.4

For the RLC circuit shown in Fig. DE6.4, suppose that $v_s(t) = 2 - 2u(t)$ V. Find $i(t)$ and $v(t)$ for $t \geq 0$.

Answer: $8te^{-4t} + 2e^{-4t}$ A; $-16te^{-4t}$ V

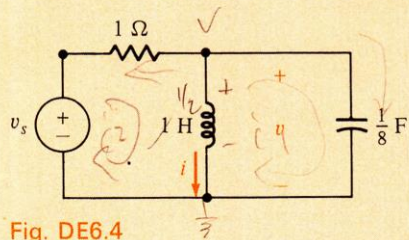


Fig. DE6.4

My copy of Bo2 is second-hand. I noticed this problem had scribbles on it. The previous owner, evidently, struggled with this problem and concluded that the value of the inductor had to be $\frac{1}{2}$ H, not 1 H as the schematic says. Symbolator can be used to show that he is right, and the book is wrong. Here is how we do it. First, we can solve the circuit as shown, with a 1 H inductor.

```
s\dc("e,2,0,2:r,1,2,1:l,1,0,1:c,1,0,1/8"):{il,vc}
{2,0}
```

```
s\only("il,vc"):s\tr("r,1,0,1:l,1,0,1,2:c,1,0,1/8,0",0):{il,vc}
```

The answer we get doesn't match the answer provided by the book. Now let's repeat the TR simulation using the $\frac{1}{2}$ H value for the inductor:

```
s\only("il,vc"):s\tr("r,1,0,1:l,1,0,1/2,2:c,1,0,1/8,0",0):{il,vc}
```

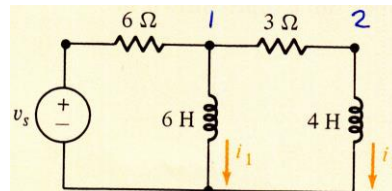
Now the answer we get matches the answer provided by the book. An expert user of Symbolator can use the **fd** gate and the **t2s** tool, which we have not discussed yet here, to find the value of the inductor given the book's answer. Below is how I found the $\frac{1}{2}$ H value myself, reverse-engineering the answer:

```
s\t2s(-16*t*e^(-4*t)):
-16/(s+4)^2
s\t2s((8*t+2)*e^(-4*t))
2*(s+8)/(s+4)^2
s\fd("r,1,0,1:l,1,0,1,2:c,1,0,1/8,0"):vc
-16*I/(I*s^2+8*I*s+8)
solve(-16/(s+4)^2=-16*I/(I*s^2+8*I*s+8),I)
I=1/2
```

The **fd** gate and **t2s** tool are used for frequency domain analysis, and will be discussed separately in a subsequent part of Symbolator's documentation.

Bo2's Drill Exercise 6.5

In the circuit below, suppose that $v_s(t) = 12 - 12 u(t)$ V. Find $i_2(t)$ for $t \geq 0$.



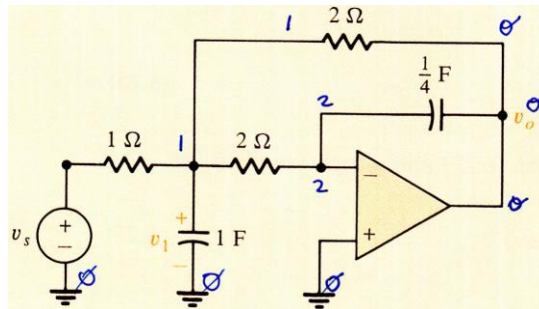
```
s\dc("e,3,0,12:r6,3,1,6:l1,1,0,6:r3,1,2,3:l2,2,0,4"):{il1,il2}
{2,0}
```

```
s\only("il2"):s\tr("r6,0,1,6:l1,1,0,6,2:r3,1,2,3:l2,2,0,4,0",0):il2
```

$$\frac{12}{11}(e^{-3t} - e^{-t/4})$$

Bo2's Drill Exercise 6.6 (Op Amp)

In the circuit below, suppose that $v_s(t) = 2 - 2u(t)$ V. Find $v_o(t)$ for $t \geq 0$.



Simulate the first interval in DC to find the initial conditions of the capacitors.

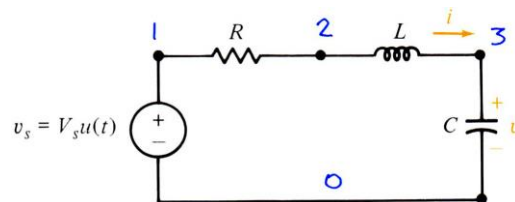
```
s\dc("e,3,0,2:r1,3,1,1:r2,1,2,2:r3,1,o,2:ca,1,0,1:
      cb,2,o,1/4:o,0,2,o"): {vca,vcb}
      {0,4}
```

Then simulate the second interval in TR, to find the voltage in node o.

```
s\only("vo"):s\tr("r1,0,1,1:r2,1,2,2:r3,1,o,2:
      ca,1,0,1,0:cb,2,o,1/4,4:o,0,2,o",0):vo
      (-4t - 4)e^{-t}
```

Bo2's Example 6.5 (Plot)

In the circuit below, use $v_s = 2/5V$, $R = 12\Omega$, $L = 2H$ and $C = 1/50F$. Find $v(t)$ and $i(t)$, and plot them for time $0 < t < 1.5$ seconds.



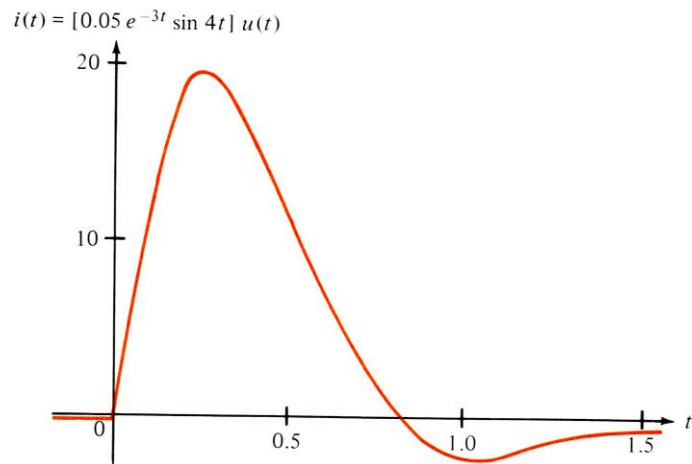
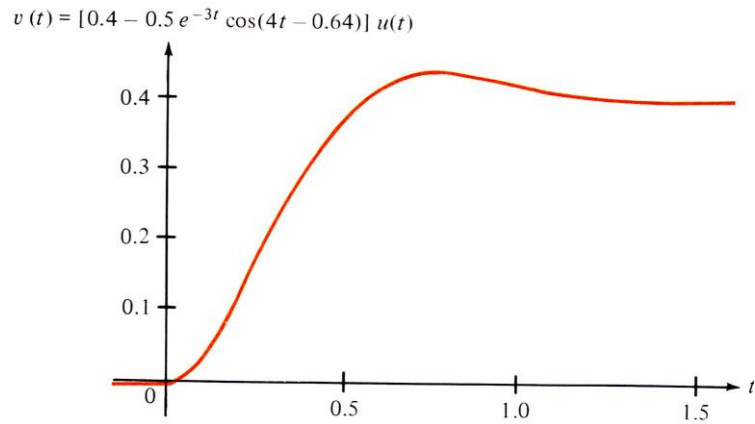
```
s\only("vc,il"):s\tr("e,1,0,2/5:r,1,2,12:l,2,3,2,0:c,3,0,1/50,0",0)
      vc
      -\frac{2}{5}e^{-3t}\cos(4t) - \frac{3}{10}e^{-3t}\sin(4t) + \frac{2}{5}
      il
      \frac{1}{20}e^{-3t}\sin(4t)
```

These are the right answers. To plot them, run the `plot` tool, thus:

```
s\plot()
```

Once the plot window opens, enter `vc` as function, `0` as minimal time and `1.5` as maximal time. Compare to the graph from the book, shown below.

Repeat the plot for `il`, and compare to the graph from the book below.



Bo2's Drill Exercise 6.7

For the circuit in Bo2's Example 6.5, use $v_s=3V$, $R=5\Omega$, $L=1/2H$ and $C=1/8F$. Find $v(t)$ and $i(t)$.

```
s\only("vc,il"):s\tr("e,1,0,3:r,1,2,5:l,2,3,1/2,0:c,3,0,1/8,0",0):{vc,il}
{ -4e^{-2t} + e^{-8t} + 3 , e^{-2t} - e^{-8t} }
```

Bo2's Drill Exercise 6.8 (Ramp)

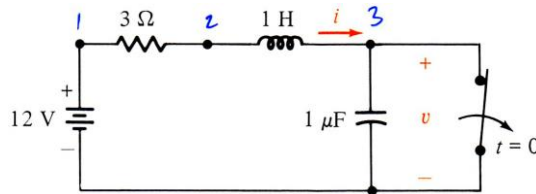
For the circuit in Bo2's Drill Exercise 6.7, find the voltage drop in the capacitor $v(t)$ if the source has a value $v_s(t) = 3 r(t)$.

```
s\only("vc"):s\tr("e,1,0,3t:r,1,2,5:l,2,3,1/2,0:c,3,0,1/8,0",0):vc
```

$$2e^{-2t} - \frac{e^{-8t}}{8} + 3t - \frac{15}{8}$$

Bo2's Example 6.6 (Plot)

For the circuit below, plot the voltage drop in the inductor for $0 < t < 5\text{ms}$.



First, find the initial conditions for $t < 0$ by running a DC simulation.

```
s\dc("e,1,0,12:r,1,2,3:l,2,3,1:c,3,0,1μ:s,3,0"):{il,vc}
{4,0}
```

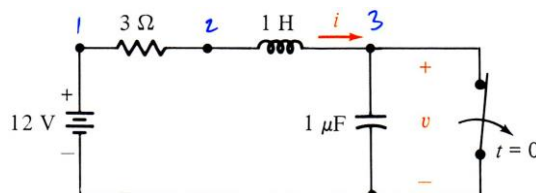
Then, find the v_l for $t \geq 0$ by running a TR simulation.

```
s\only("vl"):s\tr("e,1,0,12.:r,1,2,3:l,2,3,1,4:c,3,0,1μ,0",0)
```

Finally, run `s\plot()` and enter v_l as function, 0 as minimal time and 0.005 as maximal time. In the resulting plot you will see something outstanding: around $t = 1.57\text{ms}$, the voltage drop across the inductor is 3991 volts!

Bo2's Drill Exercise 6.9

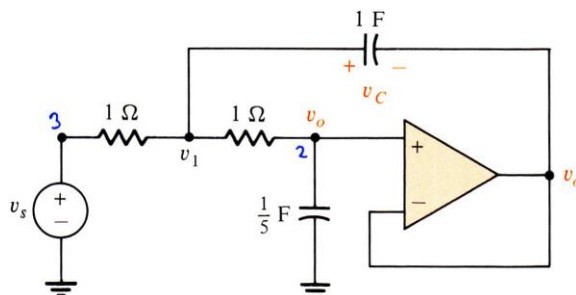
For the circuit below, $R=1\Omega$, $L=2\text{H}$, $C=1/2\text{F}$ and $i_s(t)=u(t)$. Find $i(t)$ and $v(t)$.



```
s\only("il,vc"):s\tr("j,0,1,1:r,1,0,1:l,1,0,2,0:c,1,0,1/2,0",0){il,vc}
{(-t-1)e^{-t}+1, 2te^{-t}}
```

Bo2's Figure 6.23 (Op Amp)

For the circuit below, find $v_o(t)$ if $v_s(t) = u(t)$.



`s\only("vo"):s\tr("e,3,0,1:r1,3,1,1:r2,1,2,1:ca,2,0,1/5,0:cb,1,o,1,0:o,2,o,o",0):vo`

$$1 - e^{-t} \cos(2t) - \frac{1}{2} e^{-t} \sin(2t)$$

Bo2's Drill Exercise 6.10 (Op Amp)

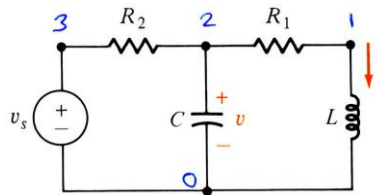
In the circuit above, change capacitor's value from 1/5 F to 25/16 F. Find $v_o(t)$ if $v_s(t) = 3 u(t)$.

`s\only("vo"):s\tr("e,3,0,3:r1,3,1,1:r2,1,2,1:ca,2,0,25/16,0:cb,1,o,1,0:o,2,o,o",0):vo`

$$-4e^{\frac{-2t}{5}} + e^{\frac{-8t}{5}} + 3$$

Bo2's Drill Exercise 6.11 (p307)

For the circuit below, suppose that $R_1 = R_2 = 1\Omega$, $L = 1H$, $C = 1F$ and $v_s(t) = 2e^{-2t} u(t)$. Find $i(t)$ if all initial conditions are zero.

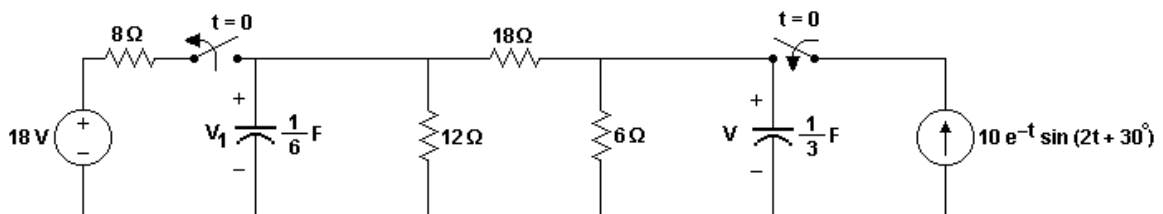


`s\only("i"):s\tr("e,3,0,2e^(-2t):r2,3,2,1:r1,2,1,1:c,2,0,1,0:l,1,0,1,0",0):i`

$$-(e^t \cos(t) - e^t \sin(t) - 1) e^{-2t}$$

A more complex problem

The circuit below comes from the circuits analysis class of Professor Eliane Boulet de Cabrera, from Universidad Tecnológica de Panama. Find v_1 and v .



Because of the switches, this is a two interval problem. The first is run in DC.

`s\dc("e,3,0,18:r8,3,1,8:ca,1,0,1/6:r12,1,0,12:r18,1,2,18:r6,2,0,6:cb,2,0,1/3"):vca,vcb`

$$\{9, 9/4\}$$

These serve as initial conditions for the second interval, which is run in TR. Since we only want v_1 and v_2 , it pays off to let Symbolator know this:

`s\only("v1,v2"):`

$\text{s}\backslash\text{tr}(\text{"ca},1,0,1/6,9:\text{r}12,1,0,12:\text{r}18,1,2,18:\text{r}6,2,0,6:\text{cb},2,0,1/3,9/4:\text{j},0,2,10\text{e}^{(-t)}\sin(2t+30^\circ)",0)$

Be patient. This took 78 seconds in my calculator, including over half a minute just to find the inverse Laplace of the two desired answers.

v1

$$\left(5\frac{\sqrt{3}}{17} - \frac{20}{17}\right)e^{-t}\cos(2t) + \left(-20\frac{\sqrt{3}}{17} - \frac{5}{17}\right)e^{-t}\sin(2t) + \left(80\frac{\sqrt{3}}{17} + \frac{193}{34}\right)e^{-\frac{t}{2}} + \left(\frac{9}{2} - 5\sqrt{3}\right)e^{-t}$$

v2

$$\left(-245\frac{\sqrt{3}}{34} - \frac{20}{17}\right)e^{-t}\cos(2t) + \left(\frac{245}{34} - 20\frac{\sqrt{3}}{17}\right)e^{-t}\sin(2t) + \left(80\frac{\sqrt{3}}{17} + \frac{193}{34}\right)e^{-\frac{t}{2}} + \left(5\frac{\sqrt{3}}{2} - \frac{9}{4}\right)e^{-t}$$

These are the correct answers, which explains why Professor Boulet – who made the problem up - is a living legend among circuit students at UTP.

Advanced use of *Expert*

The **expert** tool can be very useful in transient analysis. Using it, however, requires some knowledge. Here's two things you need to know in order to use **ex** like a boss:

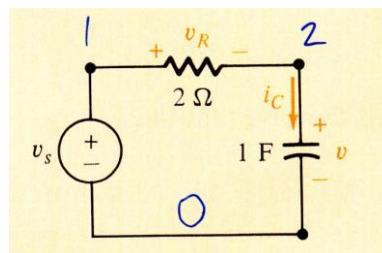
First, when Symbolator solves a problem using TR, it follows these general steps:

1. Generate a set of equations and unknowns for the system, in the frequency domain. Any time-dependent source is replaced with a dummy variable.
2. Solve these frequency domain equations.
3. Replace any dummy variable with the original source value, and convert the answers to the time domain.

Second, when Symbolator solves a problem using the expert tool, it freezes this process halfway between steps 1 and 2, so that you can tinkle with the equations and unknowns before they are solved. In the problems below, you will see that, when we tinkle with the equations, we have to do so in the frequency domain.

Bo2's Drill Exercise 4.5 (Expert)

For this circuit, we know that the capacitor's initial condition is zero, that v_s is an unknown step source (e.g. of the form $A u(t)$, where A is a constant value in volts) and that the voltage drop in the capacitor for $t > 0$ is found to be $1 - e^{-(t/2)}$. Find the voltage drop in the resistor, the current through the capacitor and the value of the source.





This problem is a match made in heaven for the **expert** tool, because we have one unknown value in the circuit (e.g. the value of the step source) and we have one known answer (e.g. the voltage drop in the capacitor.) So, the game plan here is to run this circuit through Symbulator's expert mode, add one new equation and one new unknown, and then solve. First, let's generate the new equation. A non-expert user would think that the new equation is $1 - e^{-t/2} = vc$. But you know better. You know that Symbulator solves the equations of TR problems in the frequency domain. So we have to convert this expression from the time domain to the frequency domain. Symbulator has a shortcut to invoke DiffEq's Laplace Transform to do so: `s\ts`.

$$s\ts(1 - e^{-t/2}) = vc$$

$$\frac{1}{s} - \frac{2}{2*s+1} = vc$$

We have our new equation. Copy this equation into the clipboard, since we will want to paste it in the Expert window. Now let's run the Expert simulation of the circuit. Let's define the value of the source as `a*u(t)`, since we know it's a step source; the variable `a` will serve as the unknown value, for which we will solve in the Expert mode.

$$s\lex("e,1,0,a*u(t):r,1,2,2:c,2,0,1,0")$$

When prompted, select TR as the desired option. Then you will see the typical Expert prompt. In the equations field, paste the new equation, adding the word **and** first:

$$\text{and } 1/s - 2/(2*s+1) = vc$$

In the unknown field, add the variable `a`, preceded by a comma:

$$,a$$

Let Symbulator solve the equations now. Once it is done, we want to do a quick sanity check. If the system solved correctly, we should see that Symbulator has as voltage drop in the capacitor the expression we already know from the problem statement:

$$vc$$

$$1 - e^{-\frac{t}{2}}$$

This is correct, so we proceed to ask for the answers we want:

$$ic$$

$$\frac{1}{2} e^{-\frac{t}{2}}$$

$$vr$$

$$e^{-\frac{t}{2}}$$

$$a$$

$$1$$

These are the correct answers. The value of the step source is 1 volt.

Bo2's Drill Exercise 4.13 (Expert)

For the circuit shown in Fig. DE4.13, the voltage across the $1\text{-}\Omega$ resistor is $v_R(t) = e^{-t}$ V for $t \geq 0$. Given that $v_C(0) = 0$ V, find the following functions for $t \geq 0$: (a) $i_R(t)$; (b) $v_C(t)$; and (c) $v_s(t)$.

Answer: (a) e^{-t} A; (b) $1 - e^{-t}$ V; (c) 1 V

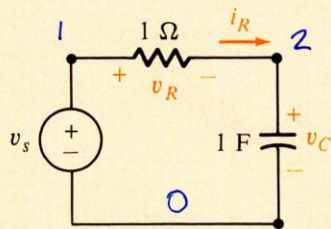


Fig. DE4.13

This problem is solved just like the one we already saw. First, find the new equation:

$$vr = s \cdot t2s(e^{-t})$$

$$vr = \frac{1}{s+1}$$

Then run the Expert simulation. I decided to use `vs` as the step value of the source:

```
slex("e,1,0,vs*u(t):r,1,2,1:c,2,0,1,0"):
```

When prompted, select TR. In the equations field, add the new equation you found:

$$\text{and } vr = 1/(s+1)$$

In the unknowns field, add the new variable:

,vs

Let Symbolator rip. Once it finishes solving, ask for the sanity check:

vr

e^{-t}

Looks good, so go ahead and ask for the answers:

ir

e^{-t}

vc

$1 - e^{-t}$

vs

1

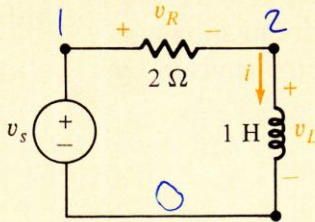
Ain't that a beauty? Yes, ma'am! Atta boy, Symbolator! Atta boy!

For the circuit shown in Fig. DE4.2, the current through the inductor is

$$i(t) = \begin{cases} 0 & \text{for } -\infty < t < 0 \\ 1 - e^{-2t} & \text{for } 0 \leq t < \infty \end{cases}$$

Find, for $t \geq 0$, (a) $v_L(t)$, (b) $v_R(t)$, (c) $v_s(t)$, and (d) the power absorbed by the inductor.

Answer: (a) $2e^{-2t}$ V; (b) $2(1 - e^{-2t})$ V; (c) 2 V; (d) $2(e^{-2t} - e^{-4t})$ W



This is the same thing as the previous two, so there is no need for commentary.

$$il = \mathcal{L}^{-1} \left\{ \frac{1}{s} - \frac{1}{s+2} \right\}$$

$$il = \frac{1}{s} - \frac{1}{s+2}$$

$$\mathcal{L}^{-1} \left(\frac{1}{s} - \frac{1}{s+2} \right) = 1 - e^{-2t}$$

Choose TR, and add ,vs to the unknowns and the following to the equations

$$\text{and } il = 1/s - 1/(s+2)$$

And solve. When done, ask for the sanity check:

$$il$$

$$1 - e^{-2t}$$

Since that adds up, ask for the answers:

$$v_l$$

$$2e^{-2t}$$

$$v_r$$

$$2 - 2e^{-2t}$$

$$v_s$$

$$2$$

$$v_l * il$$

$$2e^{-4t}(e^{2t} - 1)$$

This is the end of the TR section of the book.